

3.6: Implicit Differentiation and Related Rates

Implicit Functions

Explicit Functions

[Graph](#)

$$x^2y + y - x^2 + 1 = 0$$

$$y = \frac{x^2 - 1}{x^2 + 1}$$

$$x^2 + y^2 = 4$$

$$y = \pm\sqrt{4 - x^2}$$

$$y^3 + y^2 - xy + \frac{x^4}{4} = y$$

Implicit Differentiation:

1. Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .
2. Collect the terms with dy/dx on one side of the equation.
3. Solve for dy/dx .

Example. Find the derivatives of the following by rewriting each function explicitly before taking the derivative, and by using implicit differentiation. Compare the results.

$$y^2 = x$$

Explicitly:

$$y = \pm \sqrt{x} = \pm x^{1/2}$$
$$y' = \frac{dy}{dx} = \pm \frac{1}{2} x^{-1/2}$$
$$\frac{dy}{dx} = \boxed{\pm \frac{1}{2\sqrt{x}}}$$

Implicitly:

$$\frac{d}{dx} [y^2] = \frac{d}{dx} [x]$$
$$2y \frac{dy}{dx} = 1$$
$$\frac{dy}{dx} = \boxed{\frac{1}{2y}}$$

← equivalent →

$$\sqrt{x} + \sqrt{y} = 4$$

Explicitly:

$$\sqrt{y} = 4 - \sqrt{x} \rightarrow y = (4 - \sqrt{x})^2$$
$$\frac{dy}{dx} = 2(4 - \sqrt{x}) \left(-\frac{1}{2\sqrt{x}} \right)$$
$$= \boxed{-\frac{4 - \sqrt{x}}{\sqrt{x}}}$$

Implicitly:

$$\frac{d}{dx} [\sqrt{x} + \sqrt{y}] = \frac{d}{dx} [4]$$
$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$
$$\frac{dy}{dx} = \boxed{-\frac{\sqrt{y}}{\sqrt{x}}}$$

← equivalent →

Example. Find $\frac{dy}{dx}$ given the equation

$$\frac{d}{dx} \left[y^3 - y + 2x^3y^2 - x \right] \stackrel{d}{dx} = [8]$$

$$3y^2 \frac{dy}{dx} - 1 \frac{dy}{dx} + 6x^2y^2 + 2x^3 \frac{dy}{dx} - 1 = 0$$

$$\frac{dy}{dx} (3y^2 - 1 + 2x^3y) = 1 - 6x^2y^2$$

$$\frac{dy}{dx} = \frac{1 - 6x^2y^2}{3y^2 - 1 + 2x^3y}$$

Note: This derivative is also an implicit function

Example. Consider the equation $x^2 + y^2 = 4$.

Find $\frac{dy}{dx}$ by implicit differentiation.

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} [4]$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}}$$

Find the slope of the tangent line to the graph of the function $y = f(x)$ at the point $(1, \sqrt{3})$.

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,\sqrt{3})} = \boxed{\frac{-1}{\sqrt{3}}}$$

Find an equation of the tangent line.

$$y - y_1 = m(x - x_1)$$

↓ ↓ ↓

$$y - \sqrt{3} = \left. \frac{dy}{dx} \right|_{(1,\sqrt{3})} (x - 1)$$

$$y = -\frac{1}{\sqrt{3}}(x - 1) + \sqrt{3} = \boxed{-\frac{\sqrt{3}}{3}x + \frac{4\sqrt{3}}{3}}$$

Example. Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 4$.

$$\frac{d}{dx} \left[\frac{dy}{dx} \right] = \left[\frac{-x}{y} \right] \quad \frac{dy}{dx} = \frac{-x}{y}$$

$$\frac{d^2y}{dx^2} = \frac{y(-1) - (-x) \frac{dy}{dx}}{y^2}$$

$$\frac{d^2y}{dx^2} = \frac{y(-1) - (-x) \left(\frac{-x}{y} \right)}{y^2}$$

$$\frac{d^2y}{dx^2} = -\frac{y^2 + x^2}{y^3}$$

Related Rates:

Related rates are problems that use a mathematical relationship between two or more objects under specific constraints. From this, we can differentiate this relationship and examine how each variable changes with respect to time.

The volume of a cone with radius r and height h is given by

$$\frac{d}{dt} [V] = \frac{d}{dt} \left[\frac{1}{3} \pi r^2 h \right]$$

Find dV/dt when r and h are changing.

$$\frac{dV}{dt} = \frac{\pi}{3} \left(\frac{d}{dt} [r^2] h + r^2 \frac{d}{dt} [h] \right)$$

$$\frac{dV}{dt} = \frac{\pi}{3} \left(2r \frac{dr}{dt} h + r^2 \frac{dh}{dt} \right)$$

Find dV/dt when r is constant and h is changing.

$$\frac{dV}{dt} = \frac{\pi}{3} r^2 \frac{d}{dt} [h]$$

$$\frac{dV}{dt} = \frac{\pi}{3} r^2 \frac{dh}{dt}$$

Find dV/dt when r is changing and h is constant.

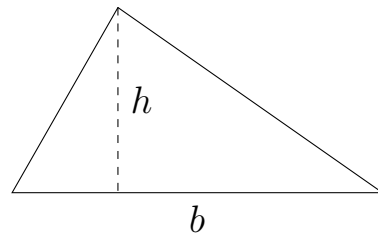
$$\frac{dV}{dt} = \frac{\pi}{3} \frac{d}{dt} [r^2] h$$

$$\frac{dV}{dt} = \frac{\pi}{3} 2r \frac{dr}{dt} h$$

Example. The altitude of a triangle is increasing at a rate of 1 cm/min while the area of the triangle is increasing at a rate of $2 \text{ cm}^2/\text{min}$. How fast is the base of the triangle changing when the altitude is 10 cm and the area is 100 cm^2 .

$$\frac{dh}{dt} = 1 \frac{\text{cm}}{\text{min}} \quad \frac{dA}{dt} = 2 \frac{\text{cm}^2}{\text{min}}$$

$$h = 10 \text{ cm} \quad A = 100 \text{ cm}^2$$



$$A = \frac{1}{2} b h \longrightarrow 100 = \frac{1}{2} b \cdot 10 \Rightarrow \boxed{b = 20}$$

$$\frac{dA}{dt} = \frac{1}{2} \left[\frac{db}{dt} h + b \frac{dh}{dt} \right]$$

$$2 = \frac{1}{2} \left[10 \frac{db}{dt} + 20 \cdot 1 \right]$$

$$4 = 10 \frac{db}{dt} + 20$$

$$\boxed{-\frac{8}{5} \frac{\text{cm}}{\text{min}} = \frac{db}{dt}}$$

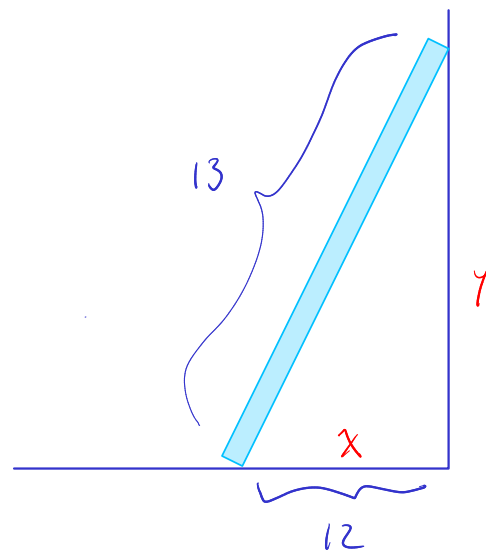
Example. The base of a 13-ft ladder leaning against a wall begins to slide away from the wall. At the instant of time when the base is 12 ft from the wall, the base is moving at a rate of 8 ft/sec. How fast is the top of the ladder sliding down the wall at that instant of time?

$$x=12 \quad \frac{dx}{dt} = 8 \text{ ft/sec}$$

$$\text{Find } \frac{dy}{dt}$$

$$x^2 + y^2 = 13^2 \quad \longrightarrow \quad y = \sqrt{13^2 - 12^2} = 5$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$



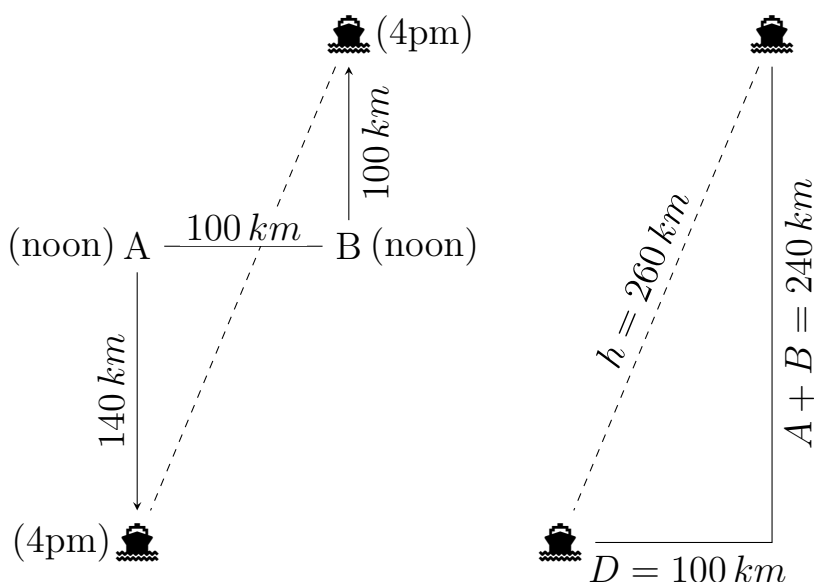
$$\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt} = -\frac{12}{5} \cdot 8 = -\frac{96}{5} = \boxed{-19.2 \text{ ft/sec}}$$

Example. At noon, ship A is 100 km west of ship B. Ship A is sailing south at 35 km/h and ship B is sailing north at 25 km/h. How fast is the distance between the ships changing at 4pm?

$$A = 140 \text{ km} \quad B = 100 \text{ km}$$

$$\frac{dA}{dt} = 35 \text{ km/h} \quad \frac{dB}{dt} = 25 \text{ km/h}$$

Find $\frac{dh}{dt}$



$$h^2 = (A+B)^2 + D^2$$

$$2h \frac{dh}{dt} = 2(A+B) \left(\frac{dA}{dt} + \frac{dB}{dt} \right) + \cancel{2D \frac{dD}{dt}}$$

$$\frac{dh}{dt} = \frac{A+B}{h} \left(\frac{dA}{dt} + \frac{dB}{dt} \right) = \frac{240}{260} (35 + 25) = \frac{12}{13} \cdot 60 \text{ km/h}$$

$$\approx 55.3846 \text{ km/h}$$